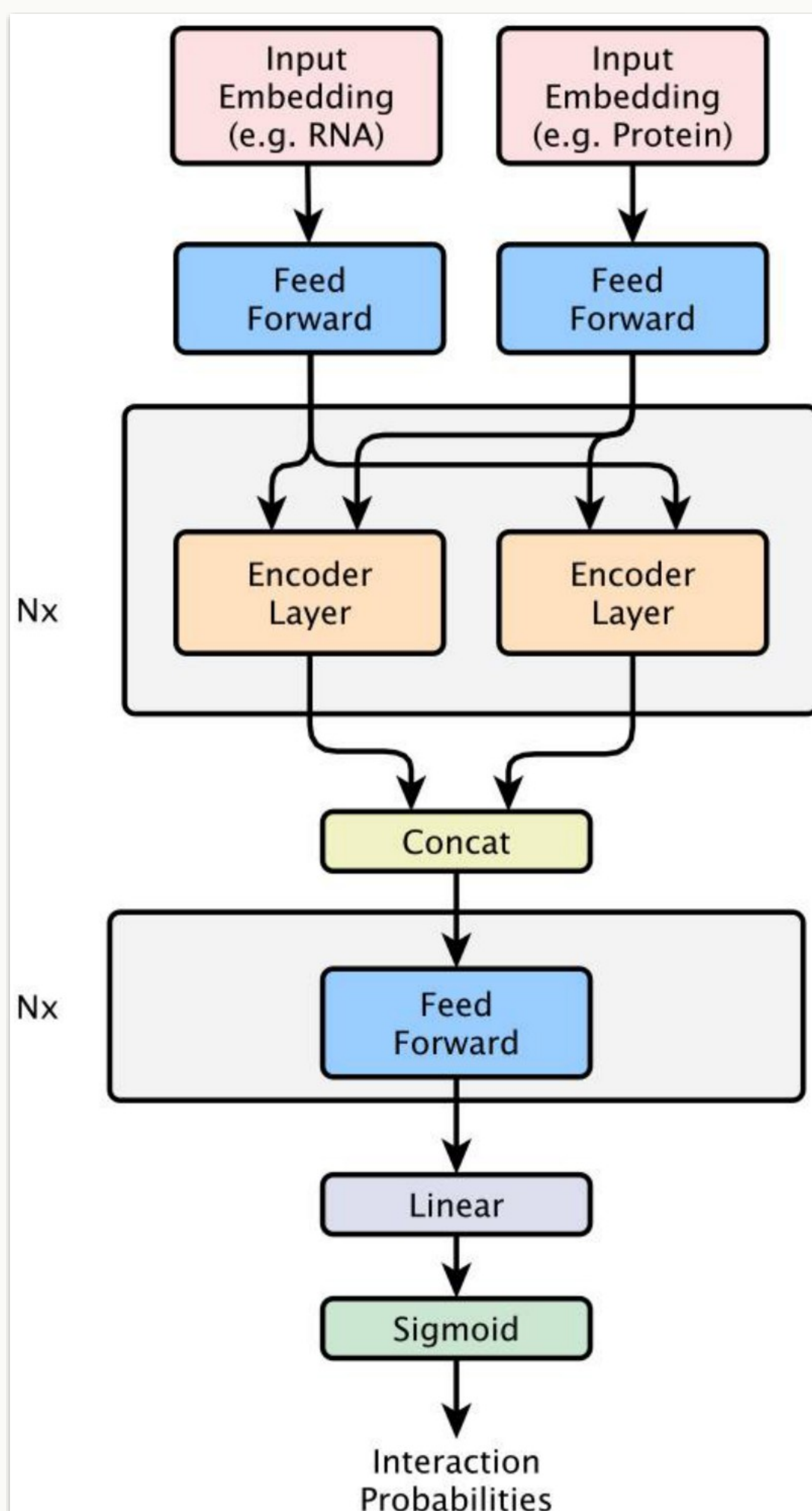


Method

$$\hat{p} = \sigma(\mathbf{W} \mathbf{h} + b)$$

RNA-FM + ESM-2 embeddings

→ symmetric encoders → sigmoid over latent $\mathbf{h}$



RPIembeddor architecture — dual embeddings
 → encoders → sigmoid. 1.4M parameters.

Problem

Predicting whether an ncRNA and protein interact **from sequence alone** is largely unsolved: most methods are protein-specific and need a dataset per protein.

Datasets exist for only a few hundred of the ~2,000 human RNA-binding proteins.

Motivation

- A **sequence-only** classifier across diverse RNA and protein types unlocks the vast interaction space without per-protein data.
- Foundation models trained on massive unlabeled corpora inject **structural and functional priors** that are otherwise unavailable for RNAs at scale.

Meta-learning across interaction types remains largely unexplored.

Dataset / Benchmark

- From **RNAInter** (>47M interactions); sequences recovered via NCBI, UniProt and Ensembl and annotated with Rfam and Pfam.
- 122,217** interactions (1:2 pos:neg) with **realistic negatives**, split by RNA family so no family is shared between train and test.
- Splits by family: TRinter **109,214** · TSfam **13,003** · external RPI2825.

RNAInter

NCBI

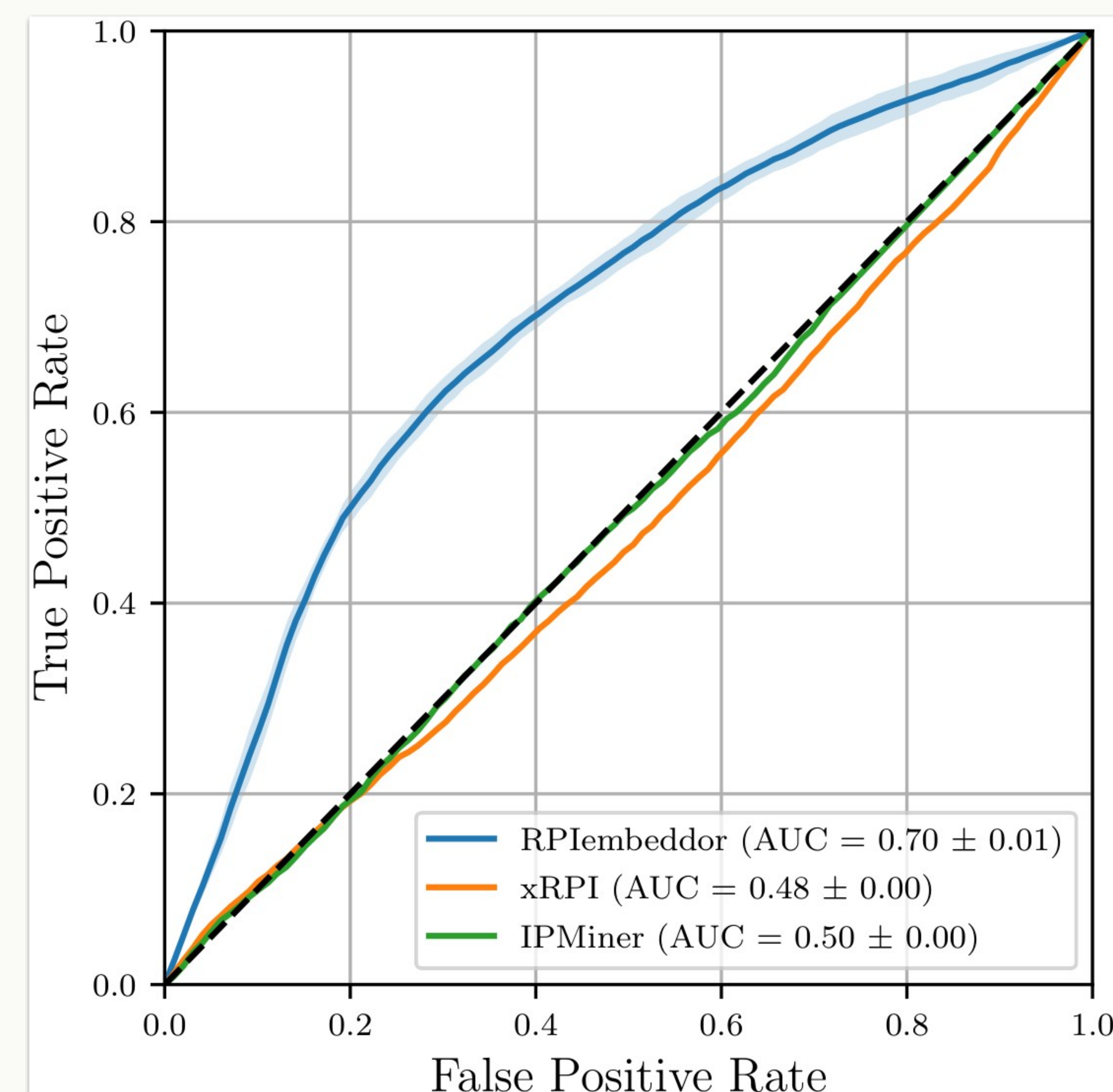
UniProt

Ensembl

Rfam

Pfam

Key Results



ROC on held-out TSfam: AUC 0.70 vs 0.48 (XRPI) and 0.50 (IPMiner), which sit at chance. RPIembeddor generalizes to unseen RNA families.

Ablation Study

EMBEDDINGS	F1	ACC
Random RNA-FM	0.0	0.624
Random ESM-2	0.0	0.624
Full (RNA-FM + ESM-2)	0.605	0.678

SO WHAT → Replace either embedding and the model collapses to a negative-only classifier — both are essential.

Headline Numbers

0.70

ROC AUC on unseen RNA families
 vs 0.48–0.50 (chance) baselines

0.586 F1 on TSfam

122K interactions

1.4M model params

Takeaway

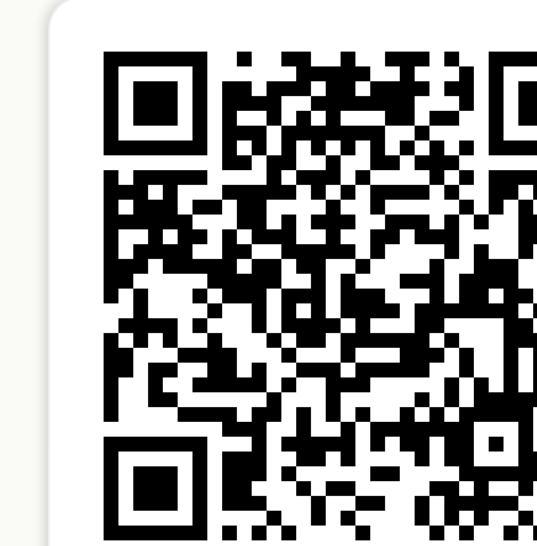
Feeding RNA + protein foundation-model embeddings into a compact attention classifier yields the first strong sequence-only predictor that generalizes across unseen RNA families.

FIRST

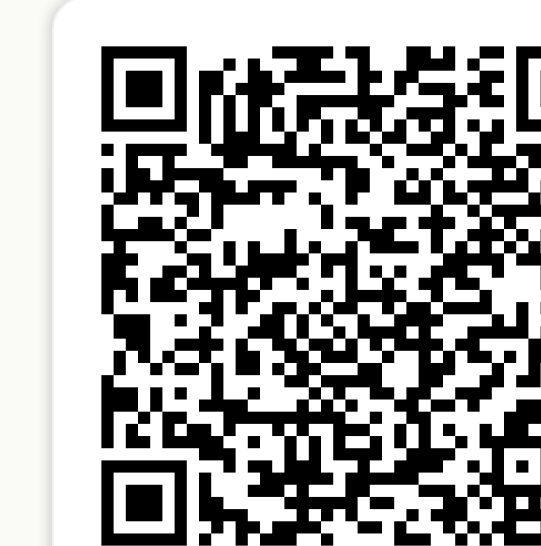
A general, sequence-only ncRNA-protein interaction predictor that needs no per-protein training data.

The released **RNAInterAct** dataset, split to remove homology bias, gives the community a fair benchmark for further progress.

Scan to Read



Paper



Dataset